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High Density Two-Dimensional Color Code

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Abstract Two-dimensional barcodes have been a topic of research for several decades. Recently, new requirements have been imposed on 2D-barcode applications, which include the capability of storing information into a small printed area. This particular requirement is specially important for applications that store cryptographic data, which needs be processed off-line. This is the case of barcodes in products like cigarettes and medicines, which store data used for validation and product verification. In this paper, we propose a 2D-colored barcode: The High Density Two-Dimensional Code (HD2DC), which is currently one of the 2D-barcodes with the highest data density. HD2DC can be generated in 8 different sizes, with 5 or 8 channel colors. To increase robustness, the system uses a Reed-Solomon error correction algorithm with 3 different levels, providing between 10% and 30% of error correction capability. We tested the HD2DC simulating two scenarios: a print-scan channel and a lossy compression stage. Results show that the proposed color barcode, besides being able to store a high density of data, is very robust to channel and compression degradations.

Keywords HD2DCC · Color barcode · 2D barcode · Reed-Solomon codes · Code design

1 Introduction

Barcodes are widely used as optical machine-readable data channel transmission mediums [1,2]. While regular barcodes store information in one dimension, 2D barcodes store information in two dimensions, enabling a higher data capacity. When compared to other technologies, like for example radio frequency tags or chips, 2D barcodes are a cheap solution for automated data transmission applications.

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Among 2D barcode techniques, the Quick Response Code (QR Code) is one of the most popular standards, being frequently used in advertisements, business cards, storefront displays, t-shirts, letter stamps, movie posters, product labels, information security, watermarking, etc. [3–7]. It was originally proposed in 1994 by the Japanese company Denso Wave Incorporated. In 2000, it was registered as the ISO/IEC 18004 standard [8]. Figure 1(a) depicts an example of a QR Code. Its basic storage information unit is the module, which carries a single information bit. A black module corresponds to a '1', while the white module corresponds to '0'.

The QR Code standard has 40 versions. The first version has modules of 21×21 pixels. For each version the number of the modules increases by 4×4, with version 40 having modules with 171×171 pixels. QR Codes use the Reed-Solomon error correction algorithm, which allows to recover a fixed amount of decoding errors. Depending on the parameters of the error correction algorithm (e.g. amount of redundancy data), different error correction rates can be achieved. QR Codes provide four different error correction rates: Low - 7%, Medium - 15%, Quality - 25%, and High - 30% [8]. Naturally, the higher the error correction rate, the lower the amount of data that can be stored in the QR Code. It is worth pointing out that the data density and the size in which the QR Code is printed affect the decoding process, causing decoding errors. In other words, for small sizes and high density levels, the QR Code decoding performance decreases considerably, requiring higher error correction levels. Although several techniques have been proposed to improve the decoding rate in these cases [9–11], this issue still limits the use of QR Codes in applications that require transmitting a higher amount of data.

Recently, several colored 2D barcodes have been proposed with the goals of improving the visual aesthetics characteristics of traditional QR Codes [12] and increasing the amount of stored bits [13–18]. An interesting example of a 2D colored barcode that uses circular dots was proposed by Shim and Kim [13]. This barcode was designed to be more robust to spatial distortions and noise, but unfortunately it can only store up to 42 bits. Another example of a colored 2D barcode is the High Capacity Color Barcode (HCCB), created by Microsoft Corporation [14]. This type of barcode uses triangular modules, with consecutive rows being separated by white lines. HCCB has two versions, one that uses 4 different colors and other that uses 8 different colors. Figure 1(b) shows an example of a HCCB barcode.

An alternative high capacity 2D barcode encoding and decoding system was proposed by Orhan Bulan and Gaurav Sharma [19]. Their scheme uses a monocrome 2D Barcode structure, considering an orientation modulation with elliptical dot arrays over the 3 channel colors (cyan, magenta, and yellow). A denser 2D Color barcode can be obtained by applying this method to the HP Color Barcode, resulting in a data density of 2,800 bytes/in^2. To improve the estimated decoded color of each module, the authors later proposed a Per-Colorant-Channel Color (PCCC) barcode that is based on the interference cancellation property [20].

More recently, Zhibo Yang *et. al.* [21] proposed a very high density 2D barcode: the HiQ. This proposal creates color information by combining n monochromatic QR Codes using 2^n colors per module. The decodification stage consists of an offline module color decision algorithm, which uses Quadratic Discriminant Analysis and Support Vector Machine learning methods to improve the resilance to errors gener-

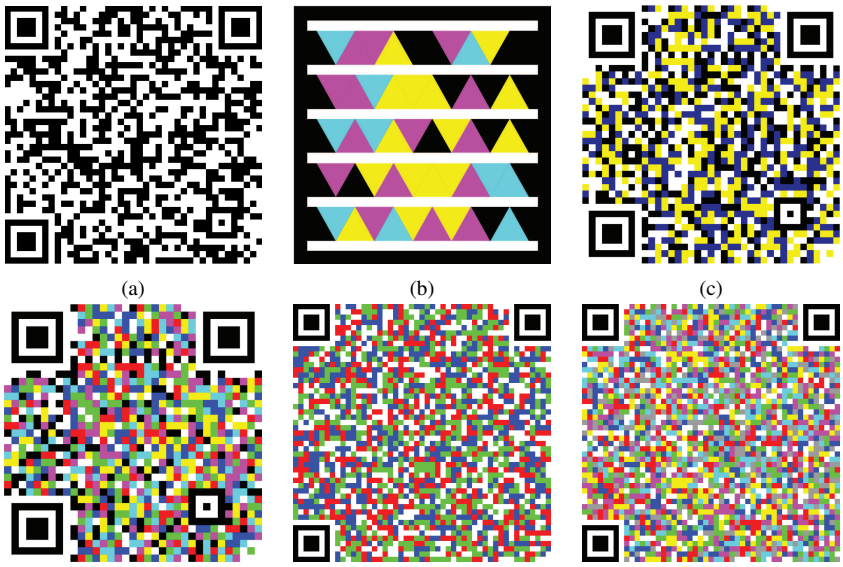


Fig. 1 Examples of barcodes: (a) regular QR Code, (b) HCCB [14], (c) HCC2D prototype for 4-colors [22], (d) HCC2D prototype for 8-colors HCC2D [22], (e) CQR Code-5 [23] and (f) CQR Code-9 [24].

ated by cross-module interference [21]. Reported results show that this denser HiQ barcode is able to store 8,097 bytes in 38×38 milimeters, which corresponds to a density of 3,617.63 bytes/in^2.This makes HiQ one of the most dense colored QR codes currently available in the literature.

Two other examples of 2D colored barcodes with fixed information capacity are: the CQR Code-5 [23] (initially named CQR Code) and the CQR-9 [24]. Examples of these two barcodes are shown in Figures 1(e) and (f), respectively. Their structure and filling direction are similar to a QR Code. The CQR Code structure is made up of modules of 49×49 pixels, which are distributed over 2 discriminate regions: the function patterns and encoding region. CQR-5 and 9 have only one size (49×49 pixels), a fixed error correction rate, and data capacities of 1,024 bits and 2,048 bits, respectively. **[Colocar a densidade deste codigo aqui, para efeitos de comparacao!]**

In this paper, we propose a new 2D colored barcode: The High Density Two-Dimensional Code (HD2DC). HD2DC is a much more flexible barcode, which allows for a choice among 3 different levels of error correction and 5 or 8 colors. In total, there are 8 possible versions of HD2C, with version 8 (8 colors and a low error correction level) reaching a data density of 4,687.09 bytes/in^2. Up to our knowledge, the HD2DC is currently one of the barcodes with the highest information density currently available. **[FALTA FALAR DAS VANTAGENS, DESVANTAGENS e CONTRIBUICOES DO HD2DC aqui!]**

This paper is divided as follows. In Sections 2 and 3, we describe the coding and decoding HD2DC algorithm. In Section 4, we analyze the impact of print-scan

channel and compression (JPEG, JPEG2k and H.264/AVC) degradations on decoding errors. Finally, in Section 5 we present our conclusions.

2 HD2DC - Data Encoding

As mentioned previously, the proposed HD2DC barcode has 8 versions. Version 1 contains modules of 30×30 pixels, version 2 contains modules of 40×40 pixels, and so on. Each version corresponds to an increase in the module size of 10×10 pixels, with version 8 having modules with 100×100 pixels. For each version, two sets of channel colors can be used: HD2DC-5 uses 5 colors (black, white, red, green and blue), while HD2DC-8 uses 8 colors (black, white, red, green, blue, cyan, magenta, and yellow). Figure 2 shows examples of HD2DCs with 5 and 8 colors, for their smaller and larger versions.

Additionally, HD2DC can have one of these 3 error correction levels: Low (10%), Medium (20%), and High (30%). The proposed HD2DC barcode uses a type of Reed-Solomon error-correction algorithm [26] known as systematic Berlekamp-Massey algorithm [27]. This algorithm uses non-binary Bose, Chaudhuri, and Hocquenghem (BCH) codes, which guarantee a maximum separation distance and a high burst error-correction capability [27]. More specifically, he HD2DC Reed-Solomon algorithm employs 16 bits per symbols and uses a primitive polynomial, given by the following equation:

$$p(D) = D^{16} + D^{12} + D^3 + 1. \tag{1}$$

A systematic Berlekamp Reed-Solomon decoder can correct up to $t = (n-k)/2$ symbols, where n is the amount of data and information symbols, k is the number of data symbols, and $(n-k)$ is the amount of redundancy symbols. The n symbols are encoded into the Encoding Region of the HD2DC. The total amount of data and redundancy symbols depend on the selected HD2DC chosen error correction level and version.

Table 1 and Table 2 show the number of data and redundancy symbols for HD2DC-5 and HD2DC-8, respectively. Notice that HD2DC-8, with a low error correction level, is able to transmit 2,906 bytes in a a printed area of 2 cm$\times$2 cm. Therefore, HD2DC-8 reaches a data density of 4,687.09 bytes/in^2. The QR Code has a maximum storage capacity (version 40, modules of 177×177 pixels) of 2,953 information bytes [8]. Therefore, when compared to the QR Code, HD2DC is more robust and can be correctly decoded even when printed on much smaller areas [4, 25].

As shown in Figure 3, the HD2DC structure is different from the QR Code and the CQR Code structures. All HD2DC versions have the same structure, which is composed of *Function Patterns* and *Encoding Region* areas. In the first step of the decoding process, Function patterns are used to detect the HD2DC barcode location. The Function patterns are divided into the following areas:

- Quiet zone: Region that is free of any markings, being located in the surrounding borders of the HD2DC barcode image.
- Finder patterns: There are 4 identical Finder Patterns in the barcode, which are located at the 4 corners of the image (upper-right, upper-left, bottom-right, and

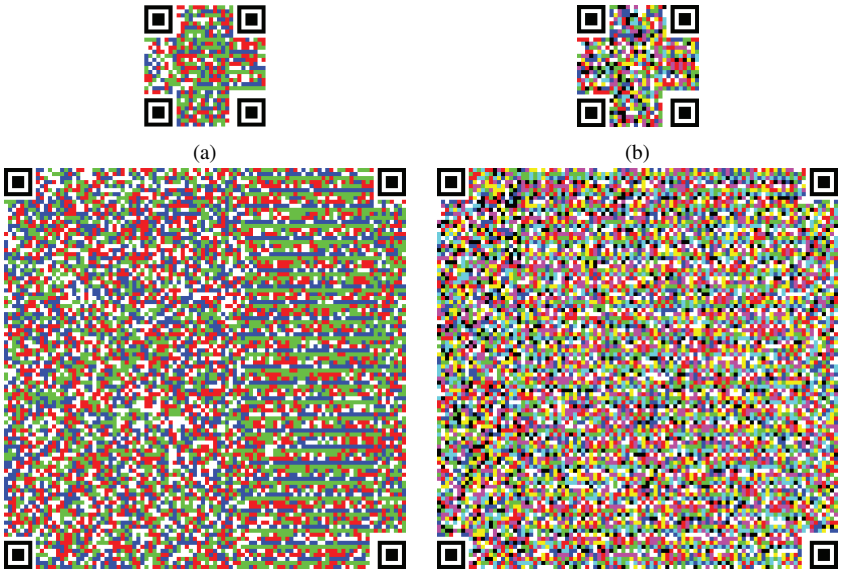


Fig. 2 Examples of HD2DC of: (a) Version 1, 5 colors, 'Low', 120 bytes, (b) Version 1, 8 colors, 'High', 90 bytes, (c) Version 8, 5 colors, 'High', 966 bytes, and (d) Version 8, 8 colors, 'Low', 2,906 bytes.

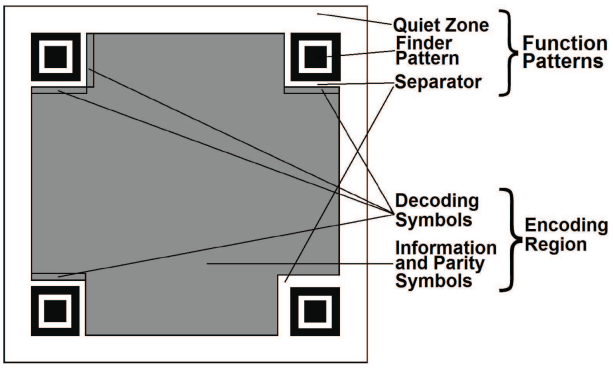


Fig. 3 HD2DC structure, showing the *Function Patterns* and *Encoding Region* areas.

bottom-left). Each one consists of 3 superimposed concentric squares, which are composed of 7×7 black modules, 5×5 white modules, and 3×3 black modules. For HD2DC-5, this is the only area where black modules are employed.

– Separators: Separators consist of white modules that are inserted between the Finder Patterns and the Encoding Region. Upper-right, upper-left, and bottom-left Finder Patterns use separators that are one-module wide, while bottom-right Finder Patterns use separators that are two-modules wide. This extra white space

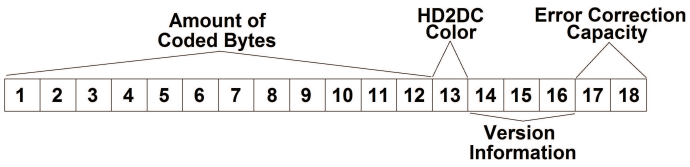


Fig. 4 HD2DC structure bits composition.

in the bottom-right Finder Patterns allows to a differentiation of this Finder Pattern and, therefore, of the orientation of the barcode.

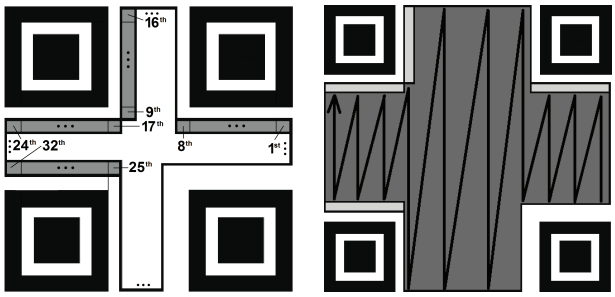
It is worth pointing out that the bottom-right Finder pattern and its two-module wide separators are features introduced in the proposed HD2DC code. More specifically, differently from previous 2D barcodes, HD2DC does not have alignment patterns or timing patterns. In other words, the modification of the fourth Finder Pattern enhances the HD2DC barcode detection robustness and, therefore, improves the storage capacity of its Encoding Region.

The Encoding region is divided into the two following areas:

- Decoding Symbols: The symbols in this area distributed in 32 modules. Each module contains 2 bits, which can be mapped as red (‘00’ bits), green (‘01’ bits), blue (‘10’ bits), and white (‘11’) module colors. These 32 modules are shown in the gray area in Figure 5(a). From the 64 available bits, 63 bits are reserved to the BCH(63,18) bits [27] and 1 bit is set as ‘0’ (remainder bit). As shown in Figure 4, in BCH(63,18), from the 18 bits that carry the HD2DC structure information, 12 bits specify the number of coded bytes (1 to 2,906 bytes), 1 bit defines the Encoding Region color (HD2DC-5 or HD2DC-8), 3 bits specify the version (version 1 to 8), and 2 bits specify for the error correction capacity (Low, Medium, or High). Figure 5(a) shows an enlarged image of the HD2DC structure.
- Information and Parity Symbols: This area is filled with Berlekamp Reed-Solomon [28] information and parity symbols. The bits corresponding to information and redundancy symbols are placed in the dark gray modules of the encoding region, following the vertical bottom-up direction, from the most right to the most left column, as illustrated in Figure 5(b). The error correction polynomial (Eq. 1) and the quantity of bits per symbol ($s = 16$) are the same used by CQR Codes [23, 24]. Considering that HD2DC has many versions and error correction levels, there is a total of 48 possible HD2DC architectures (24 for each HD2DC color).

As mentioned earlier, the bottom-right Finder Pattern and its two-modules wide separators are new features that eliminate the need for alignment patterns or timing patterns.

Tables 1 and 2 also show the HD2DC construction architectures for HD2DC version 5 (HD2DC-5) and version 8 (HD2DC-8), respectively. **(SER MAIS ESPECIFICO! NESTA TABELAS TEM os PARAMETROS. DIZER QUE PARAMETROS SAO ESTES! Esta frase estaria adequada se fosse uma figura.)** The colors used in HD2DC-5 (red, green, blue and white) and HD2DC-8 (red, green, blue, cyan, magenta, yellow, white and black) were chosen because of their maximum separation



(a) (b)

Fig. 5 HD2DC modules: (a) BCH(63,18) distribution, and (b) HD2DC filling direction.

Table 1 HD2DC-5 color information structure.

Version	Encoding Region Modules	*n*	*k*	*t*	Error Correction %	Information Bytes	Remainder Bits/Modules
1	595	74	60	7	9.46%	120	6/3
1	595	74	46	14	18.92%	92	6/3
1	595	74	30	22	29.73%	60	6/3
2	1,295	161	129	16	9.94%	258	14/7
2	1,295	161	97	32	19.88%	194	14/7
2	1,295	161	65	48	29.81%	130	14/7
3	2,195	274	220	27	9.85%	440	6/3
3	2,195	274	166	54	19.71%	332	6/3
3	2,195	274	110	82	29.93%	220	6/3
4	3,295	411	329	41	9.98%	658	14/7
4	3,295	411	247	82	19.95%	494	14/7
4	3,295	411	165	123	29.93%	330	14/7
5	4,595	574	460	57	9.93%	920	6/3
5	4,595	574	346	114	19.86%	692	6/3
5	4,595	574	230	172	29.97%	460	6/3
6	6,095	761	607	77	10.12%	1,214	14/7
6	6,095	761	455	153	20.11%	910	14/7
6	6,095	761	303	229	30.09%	606	14/7
7	7,795	974	778	98	10.06%	1,556	6/3
7	7,795	974	584	195	20.02%	1,168	6/3
7	7,795	974	388	293	30.08%	776	6/3
8	9,695	1,211	967	122	10.07%	1,934	14/7
8	9,695	1,211	725	243	20.07%	1,450	14/7
8	9,695	1,211	483	364	30.06%	966	14/7

in the Red, Green, Blue (RGB) and Hue, Saturation, Value (HSV) color spaces, as shown in Figures 6(a) and (b). When 5 colors are used, each module of the Encoding Region contains the following color codes: red ‘00’, green ‘01’, blue ‘10’ and white ‘11’. For 8 color options, colors are red ‘000’, green ‘001’, blue ‘010’, cyan ‘011’, magenta ‘100’, yellow ‘101’, white ‘110’, and black ‘111’. When the maximum amount of bytes is not reached, the remaining bits are filled with pseudo-random values.

Table 2 HD2DC-8 color information structure.

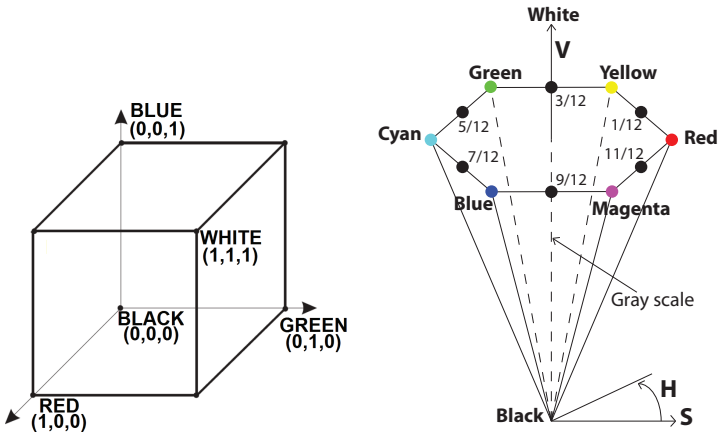
Version	Encoding Region Modules	n	k	t	Error Correction %	Information Bytes	Remainder Bits/Modules
1	595	111	89	11	9.91%	178	9/3
1	595	111	67	22	19.82%	134	9/3
1	595	111	45	33	29.73%	90	9/3
2	1,295	242	194	24	9.92%	338	13/4.33
2	1,295	242	146	48	19.83%	292	13/4.33
2	1,295	242	98	72	29.75%	196	13/4.33
3	2,195	411	329	41	9.98%	658	9/3
3	2,195	411	247	82	19.95%	494	9/3
3	2,195	411	165	123	29.93%	330	9/3
4	3,295	617	495	61	9.89%	990	13/4.33
4	3,295	617	371	123	19.94%	742	13/4.33
4	3,295	617	247	185	29.98%	494	13/4.33
5	4,595	861	689	86	9.99%	1,378	9/3
5	4,595	861	517	172	19.98%	1,034	9/3
5	4,595	861	345	258	29.97%	690	9/3
6	6,095	1,142	912	115	10.07%	1,824	13/4.33
6	6,095	1,142	684	229	20.05%	1,368	13/4.33
6	6,095	1,142	456	343	30.04%	912	13/4.33
7	7,795	1,461	1,167	147	10.06%	2,334	9/3
7	7,795	1,461	875	293	20.05%	1,750	9/3
7	7,795	1,461	583	439	30.05%	1,166	9/3
8	9,695	1,817	1,453	182	10.02%	2,906	13/4.33
8	9,695	1,817	1,089	364	20.03%	2,178	13/4.33
8	9,695	1,817	725	546	30.05%	1,450	13/4.33

To generate the Berlekamp Reed-Solomon symbols (with $s = 16$ bits), k symbols are used (from information bytes) as input and n symbols are created as output of the error correction algorithm. The n symbols are mapped into $n \times s$ bits located into the Information and Parity Symbols area, according to the filling direction shown in Figure 5(b). Naturally, version 8 reaches a higher data capacity since each of its module pixel is able to store 3 bits, while the module pixels in version 5 are able to store X bits [15,29]. **[E a versao 5 pode armazenar quantos bits por pixel?]**

3 HD2DC - Data Decoding

To decode a HD2DC, we first capture the barcode image using a digital camera. Then, the decoding algorithm locates the 4 Finder Patterns and rotates the HD2DC image to the 'ideal' position (see Figures 2, 3, and 5). For each R, G, and B channels, thresholds th_1, th_2 and th_3 are estimated using Otsu's segmentation algorithm [31]. Using Canny edge detection algorithm [32], the center of the 4 Finder Patterns are obtained by calculating the ratio of black and white modules in the Function Patterns.

To estimate the version of the HD2DC barcode, we first compute the mean size (in pixels) of the modules around the 4 Finder Patters of the barcode, S_{FP}. Next, for each version ($1 \geq i \geq 8$), we calculate the mean module size (in pixels), S_M, computed



(a) (b)

Fig. 6 Color space constellation for used encoding and decoding processes: (a) RGB Cube and (b) HSV with normalized boundaries between HD2DC selected colors.

as:

$$S_M(i) = \frac{0.25 \cdot (D_{AB} + D_{BC} + D_{CD} + D_{DA})}{13 + 10 \cdot i}, \tag{2}$$

where D_{XY} corresponds to the distance (in pixels) between the consecutive Finder Patterns X and Y, with the 4 Finder Patterns being A, B, C, and D. Finally, the version is estimated using the following equation:

$$\text{Version} = \begin{cases} 1, & \text{if } \frac{S_M(1)+S_M(2)}{2} < S_{FP}; \\ 2, & \text{if } \frac{S_M(2)+S_M(3)}{2} < S_{FP} \leq \frac{S_M(1)+S_M(2)}{2}; \\ 3, & \text{if } \frac{S_M(3)+S_M(4)}{2} < S_{FP} \leq \frac{S_M(2)+S_M(3)}{2}; \\ 4, & \text{if } \frac{S_M(4)+S_M(5)}{2} < S_{FP} \leq \frac{S_M(3)+S_M(4)}{2}; \\ 5, & \text{if } \frac{S_M(5)+S_M(6)}{2} < S_{FP} \leq \frac{S_M(4)+S_M(5)}{2}; \\ 6, & \text{if } \frac{S_M(6)+S_M(7)}{2} < S_{FP} \leq \frac{S_M(5)+S_M(6)}{2}; \\ 7, & \text{if } \frac{S_M(7)+S_M(8)}{2} < S_{FP} \leq \frac{S_M(6)+S_M(7)}{2}; \\ 8, & \text{if } S_{FP} \leq \frac{S_M(7)+S_M(8)}{2}. \end{cases} \tag{3}$$

Once the version is estimated and validated with the Encoding Region color bit (see Figure 4), we can correct the perspective distortion [33]. Due to the fourth Finder Patter, the projective transform can be performed using the 4 Finder Pattern centers (matching points: A, B, C, and D). The planar projective transform is a linear transformation of 3 homogeneous vectors, which are represented by a non-singular 3×3 matrix, given by the following equation:

$$x' = \begin{bmatrix} x'_1 \\ x'_2 \\ x'_3 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = Hx. \tag{4}$$

In its inhomogeneous form, Eq. 4 can be written as:

$$x' = \frac{x'_1}{x'_3} = \frac{h_{11} \cdot x + h_{12} \cdot y + h_{13}}{h_{31} \cdot x + h_{32} \cdot y + h_{33}}, \tag{5}$$

and

$$y' = \frac{x'_2}{x'_3} = \frac{h_{21} \cdot x + h_{22} \cdot y + h_{23}}{h_{31} \cdot x + h_{32} \cdot y + h_{33}}. \tag{6}$$

Each Finder Pattern point generates two linear equations, which are:

$$x' \cdot (h_{31} \cdot x + h_{32} \cdot y + h_{33}) = h_{11} \cdot x + h_{12} \cdot y + h_{13} \tag{7}$$

and

$$y' \cdot (h_{31} \cdot x + h_{32} \cdot y + h_{33}) = h_{21} \cdot x + h_{22} \cdot y + h_{23}. \tag{8}$$

Therefore, 8 linear equations are created using the 4 Finder Patterns positions that are used as inputs to Equations 7 and 8. This is sufficient to find the transformation matrix H and perform the projective transform.

The projective transformation matrix eliminates geometrical distortions that may be introduced during printing and capturing the HD2DC image. After performing the projective transform, the correct rotation of the 4 Finder Patterns is computed and we can recover the Finder Pattern with a two-modules wide separator (see Figure 3), which should be located on the bottom-right corner of the barcode. Figure 8(b) illustrates an example of a HD2DC after applying the projective transform. Notice that the HD2DC decoding does not require extra Function Patterns, like alignment patterns or timing patterns, as used by QR Codes and HCC2D.

The next step consists of segmenting the colors of the captured barcode image. Color segmentation is performed in the HSV color space. It is worth noting that the HSV is a color space that is more closely aligned to the way human viewers perceive colors [30]. In this color space, colors are represented by their Hue (H), Saturation (S), and Value (V) characteristics. As shown in Figure 6(b), in HSV the different hues of colors are arranged in a radial slice, around a central axis of neutral colors which ranges from black to white. So, the Hue value of a specific color is given by the angle ($360 < H \leq 0$) of the corresponding hue value. The Saturation corresponds to the purity of the color and is expressed by the distance to the gray scale axis of the color space (central axis in Figure 6(b)), with more pure (saturated) colors being represented at a maximum distance (see red, green, blue, cyan, yellow, and magenta points in the figure) and less saturated colors being represented closer to the gray scale axis. The Value of a color corresponds to the luminance intensity (achromatic) information.

The RGB captured image is first converted to the HSV color space:

$$I(i,j,c) = [H(i,j), S(i,j), V(i,j)]. \tag{9}$$

where i and j are the spatial coordinates of the image and c corresponds to the color channel. For brevity, we omit here the equations to convert from RGB to HSV, which can be found in the classic image processing book by Gonzales and Woods [30].

After the image is converted to HSV, the maximum and minimum values of the Value component (V) are computed:

$$V_{max} = \max\{V(i,j)\} \tag{10}$$

and

$$V_{min} = \min\{V(i,j)\}. \tag{11}$$

Then, using the following equation, a color segmentation is performed to retrieve the decoding symbols and the HD2DC parameters:

$$\text{HD2DC-5}(i,j) = \begin{cases} \text{White}, & \text{if } V(i,j) > \frac{(3 \cdot V_{max} + V_{min})}{4}; \\ \text{Red}, & \text{if } -60^\circ < H(i,j) \leq 60^\circ; \\ \text{Green}, & \text{if } 60^\circ < H(i,j) \leq 180^\circ; \\ \text{Blue}, & \text{if } 180^\circ < H(i,j) \leq 300^\circ. \end{cases} \tag{12}$$

If the HD2DC barcode has 5-colors, there is no need to perform a second segmentation. On the other hand, if this is a barcode with 8 colors, an additional 8-colors segmentation is performed for each pixel, as given by:

$$\text{HD2DC-8}(i,j) = \begin{cases} \text{White}, & \text{if } V(i,j) > \frac{(3 \cdot V_{max} + V_{min})}{4}; \\ \text{Black}, & \text{if } V(i,j) > \frac{(3 \cdot V_{min} + V_{max})}{4}; \\ \text{Red}, & \text{if } -30^\circ < H(i,j) \leq 30^\circ; \\ \text{Yellow}, & \text{if } 30^\circ < H(i,j) \leq 90^\circ; \\ \text{Green}, & \text{if } 90^\circ < H(i,j) \leq 150^\circ; \\ \text{Cyan}, & \text{if } 150^\circ < H(i,j) \leq 210^\circ; \\ \text{Blue}, & \text{if } 210^\circ < H(i,j) \leq 270^\circ; \\ \text{Magenta}, & \text{if } 270^\circ < H(i,j) \leq 330^\circ. \end{cases} \tag{13}$$

After the colors of each module are determined, the bits are extracted from the Encoding Region, forming a Berlekamp Reed-Solomon sequence of symbols [28]. The Berlekamp Reed-Solomon decoding algorithm is applied and (possible) errors are corrected. If the decoded sequence of symbols has more than t errors, it will not be possible to recover all the information.

Figure 7 shows a flowchart of the decoding steps presented in this section. Table 3 shows the information data density for each version of the HD2DC barcode, while Table 4 [15, 22] shows a comparison of maximum data density (in bytes/in^2) of HD2DC and currently available 2D Colored barcodes. Notice, currently, HD2DC is the colored barcode with the highest data density.

4 Experimental Results

4.1 Print-Scan Channel

To test the proposed scheme as a print-scan channel, we print HD2DC images and capture them using the mobile phones Samsung Galaxy S5 and Samsung Galaxy J5,

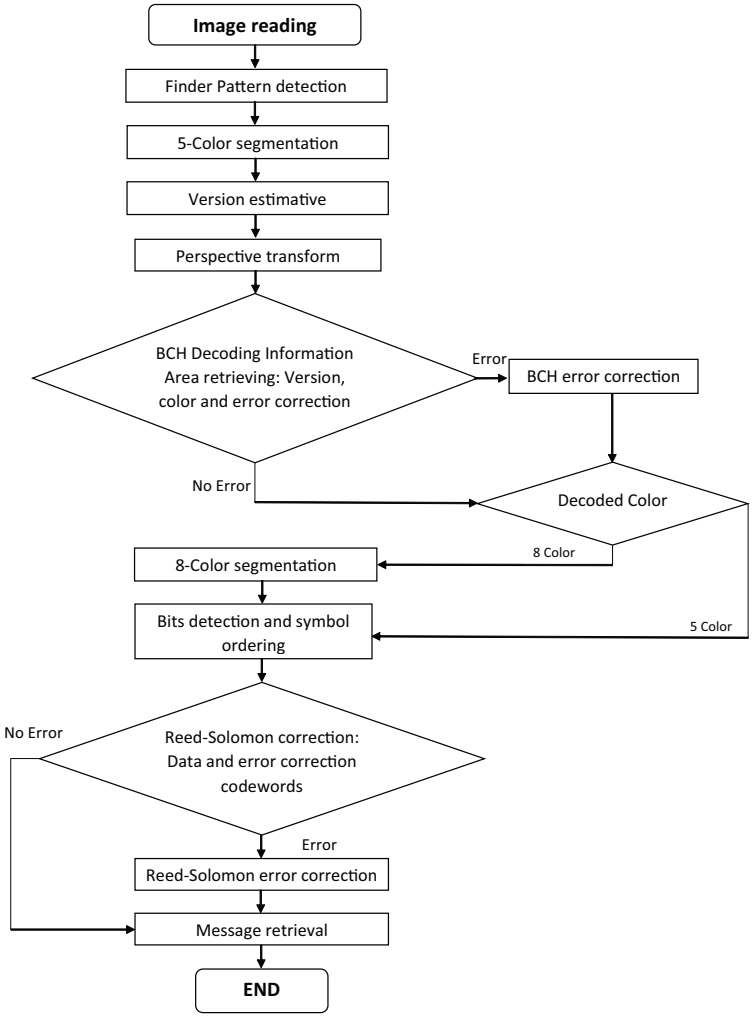


Fig. 7 Flowchart of HD2DC decoding process.

which have 16 and 8 Megapixels built-in cameras, respectively. The image database built for our tests is composed of 14 snapshots of 3 different HD2DC barcodes, considering each HD2DC version. In total, we generated 672 barcode images with 600 dpi and a module size of 0.2 mm $\times$ 0.2 mm, which is the smallest size that can be correctly decoded (see Table 3). A laser printer Xerox Color 560 was used to print the HD2DC barcode images on regular office A4 paper with 120 g/m^2 density.

We measured the percentage of corrupted symbols versus the distance (in centimeters) between the mobile and the printed barcode image. Figure 8(a) illustrates an example of a printed version 3 HD2DC-8 barcode, while Figure 8(b) illustrates its rotated and processed version. In this figure, there are visible JPEG-compression

Table 3 Sizes and information bits density of HD2DC printed at 600 dpi.

Version	HD2DC Color	HD2DC Area (in^2)	Information bytes (L)	Data Density (bytes/in^2)
1	5	0.0558	120	2,150.53
2	5	0.0992	258	2,600.80
3	5	0.1550	440	2,838.70
4	5	0.2232	658	2,948.02
5	5	0.3038	920	3,028.30
6	5	0.3968	1,214	3,059.47
7	5	0.5022	1,556	3,098.36
8	5	0.6200	1,934	3,119.35
1	8	0.0558	178	3,189.96
2	8	0.0992	338	3,407.25
3	8	0.1550	658	4,245.16
4	8	0.2232	990	4,435.48
5	8	0.3038	1,378	4,535.87
6	8	0.3968	1,824	4,596.77
7	8	0.5022	2,334	4,647.55
8	8	0.6200	2,906	4,687.09

Table 4 Density of 2D Barcodes printed at 600 dpi [15,16,22,19,20]. **Melhorar esta caption.**

2D Colored Barcode	Data Density (bytes/in^2)
HD2DC	4,687.09
HCC2D **[?].**	3,819.76
HiQ **[?].**	3,617.63
PCCC **[?].**	2,800.00
High Capacity 3D Print Codes **[?].**	2,500.00
HCCB **[?].**	2,000.00
QR Code **[?].**	627.00

and halftoning artifacts (more visible on cyan modules), which impact the symbol decoding rates.

Average decoding results for Versions 1, 3, 6, and 8 are shown in Figures 9-12. In these graphs, snapshots resulting in a percentage of corrupted symbols above the magenta dotted line cannot be decoded, while percentages between cyan and magenta dotted lines can only be decoded for the High error correction level. Percentages between the yellow and cyan dotted lines can be decoded with a Medium (or a High) error correction level. Finally, percentages of corrupted symbols under the yellow dotted line can be decoded with either a Low, Medium, or High error correction levels. As seen in Table 3, decoded HD2DC barcodes with Low error correction levels obtain the highest data density.

All printed HD2DC versions (1 to 8) were correctly decoded. Depending on the version, both the Samsung Galaxy S5 and J5 were able to correctly decode barcodes with distances ranging from 7 cm to 12 cm. Nevertheless, results show that the quality



(a) (b)

Fig. 8 HD2DC Version 5, 8-colors, and High error correction level: (a) image snapshot captured at a distance of 7 cm from the camera (Samsung Galaxy J5) and (b) rotated and processed version of (a).

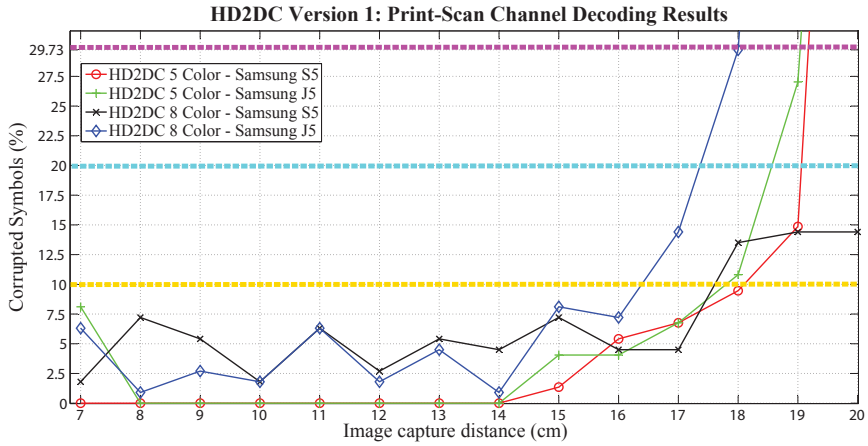


Fig. 9 HD2DC Version 1 decoding results due capture distance (7 cm to 20 cm) on Samsung Galaxy S5 and J5.

of the camera affects the percentage of corrupted symbols. For all HD2DC versions, the Samsung Galaxy S5 camera presented a better performance than the Samsung Galaxy J5 camera, for both HD2DC-5 and HD2DC-8 versions.

4.2 Robustness to Compression

Given that compression algorithms may introduce degradations in the captured barcode image and, therefore, incur in decoding errors [34], we tested the robustness of the proposed barcode to compression. To perform these tests, we used three popular lossy compression algorithms: JPEG [35], JPEG2k [36, 37], and H.264/AVC [38–40]. JPEG was chosen because it is the most popular image compression standard, while JPEG2k was chosen because it provides higher compression rates. On the other hand,

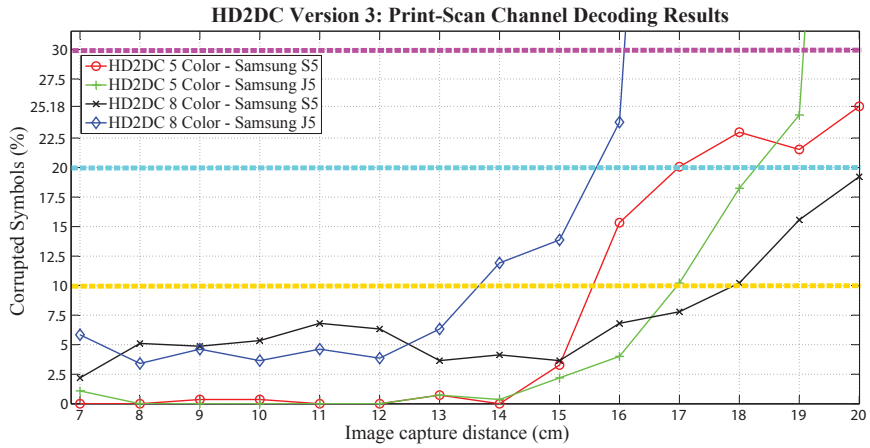


Fig. 10 HD2DC Version 3 decoding results due capture distance (7 cm to 20 cm) on Samsung Galaxy S5 and J5.

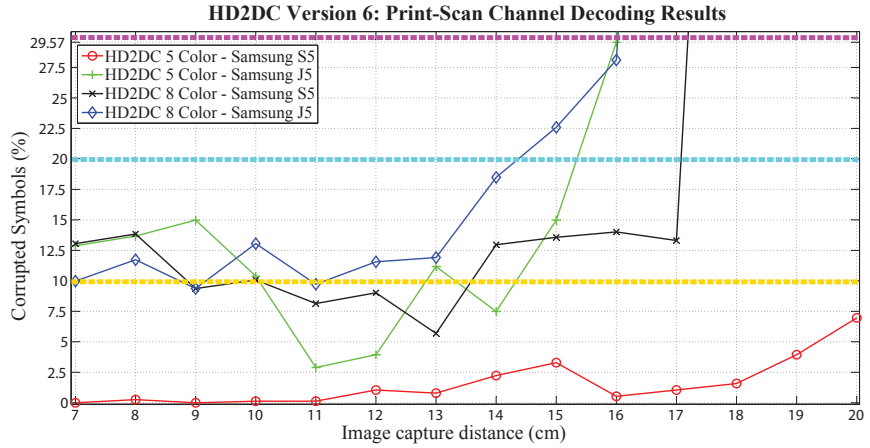


Fig. 11 HD2DC Version 6 decoding results due capture distance (7 cm to 20 cm) on Samsung Galaxy S5 and J5.

although H.264/AVC is a video compression standard, it was chosen because it has been used for image compression with very good results [41,42].

Our tests were performed with the standard structure of HD2DC, which includes regular 4 modules of quiet zone on each side. The original HD2DC images (1 module per pixel) were resized, using the nearest neighbor algorithm, making each module 10×10 pixels. Figures 2(b) and (c) show a sample of the 240 HD2DC barcode images used in our tests. For a better analysis of the symbol corruption rate due to compression, we use only the HD2DCs with a high error correction level.

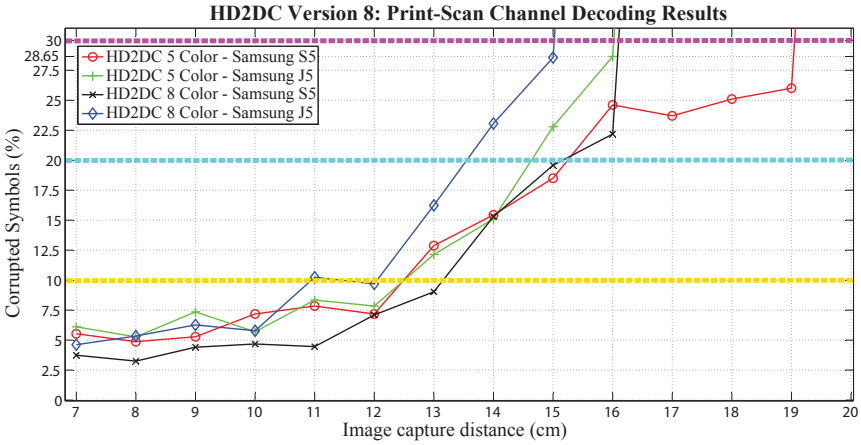


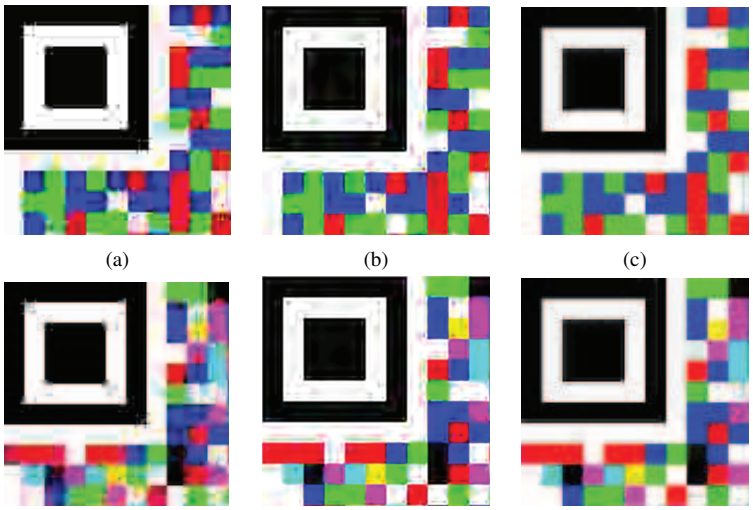
Fig. 12 HD2DC Version 8 decoding results due capture distance (7 cm to 20 cm) on Samsung Galaxy S5 and J5.

An analysis of the compression bitrate values (bits per pixel - bpp) versus the percentage of the total number of corrupted symbols. Figures 13(a)-(c) depict examples of HD2DC-5 compressed using JPEG, JPEG2k, and H.264/AVC, while Figures 13(d)-(f) depict examples of HD2DC-8 compressed with JPEG, JPEG2k and H.264/AVC. The compression bitrates used in this example are around 0.48 bpp for HD2DC-5 and around 0.40 bpp for HD2DC-8. For a better visualization of the compression visual degradations, the images show an enlarged area around the upper left Finder Pattern.

Our tests were performed using 5 different HD2DCs images for each color, version, and compression algorithm. HD2DC compressed images were processed until their maximum decodable compression rate. Notice that, there is a big difference between the images in Figures 13(a)-(c) and Figures 13(d)-(f), which correspond to HD2DC-5 and HD2DC-8, respectively. As expected, the image compressed with JPEG has the worst quality (a higher number of visible degradations around finder patterns), while the image compressed with H.264/AVC has the best quality (fewer number of visible degradations).

Figures 15-18 show the average percentage of corrupted symbols versus the bitrate for versions 1, 3, 6, and 8, respectively. Each graph contains the results obtained for the three compression algorithms. These results are consistent with the qualitative results presented in Figure 13. Figure 14(a), (b) and (c) depict a zoom area of the barcode images corresponding to the maximum decodable JPEG, JPEG2k and H.264/AVC bitrates for HD2DC-8 Version 8.

In the case of the JPEG, HD2DC-5 and HD2DC-8 can be decoded with bitrates greater than 0.4156 bpp and 0.4151 bpp, respectively, which correspond to the highest JPEG compression rates (level 100 in JPEG compression) obtained for these images. At these bitrates, the HD2DC barcodes present an average symbol error of 27.93%. Since the Reed-Solomon algorithm can correct this percentage of corrupted symbols



(a) (b) (c)

(d) (e) (f)

Fig. 13 Enlarged HD2DC of 5 and 8 colors on Version 4, compressed with: (a) JPEG@0.4908 bpp and 17.03% corrupted symbols, (b) JPEG2k@0.4720 bpp and 2.92% corrupted symbols, (c) H.264/AVC@0.4727 bpp and 0.97% corrupted symbols, (d) JPEG@0.4951 bpp and 27.93% corrupted symbols, (e) JPEG2k@0.4002 bpp and 1.80% corrupted symbols, finally (f) H.264/AVC@0.3929 bpp and 0.00% corrupted symbols.



(a) (b) (c)

Fig. 14 Enlarged 8-colors HD2DC of Version 8, compressed with: (a) JPEG@0.4654 bpp and 25.92% corrupted symbols, (b) JPEG2k@0.1106 bpp and 29.50% corrupted symbols, finally (c) H.264/AVC@0.5725 bpp and 1.82% corrupted symbols.

for a High error correction level, the HD2DC barcodes are perfectly decoded for JPEG bitrates greater than 0.4151 bpp.

For JPEG2k, HD2DC-5 and HD2DC-8 can be decoded with birates greater than 0.0816 bpp and 0.1169 bpp, respectively, which correspond to JPEG2k compression levels 285 and 200. For these bitrates, the barcode have an average percentage of corrupted symbols equal to 26.31% and 28.67%, respectively. Therefore, these information symbols can be corrected if a High error correction level is used. Results for JPEG2k show how the color space can affect the percentage of corrupted symbols. For the same compression parameters, HD2DC-8 has a higher percentage of corrupted symbols than HD2DC-5.

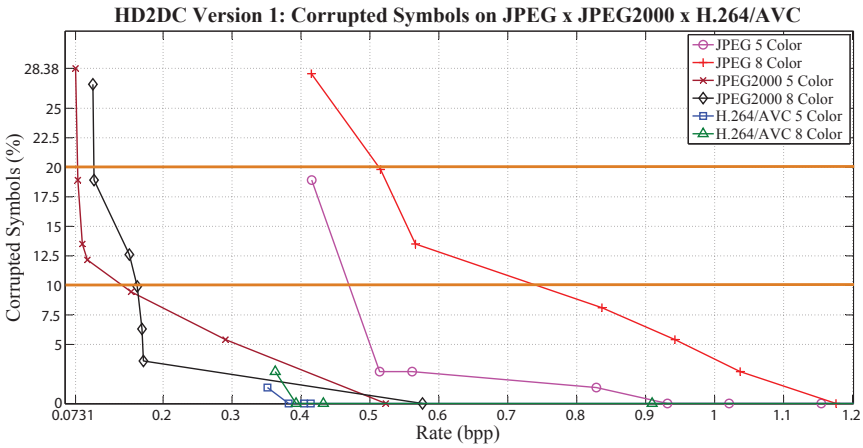


Fig. 15 Percentage of corrupted symbols of HD2DC Version 1 versus compression bitrate (0-1.2 bpp).

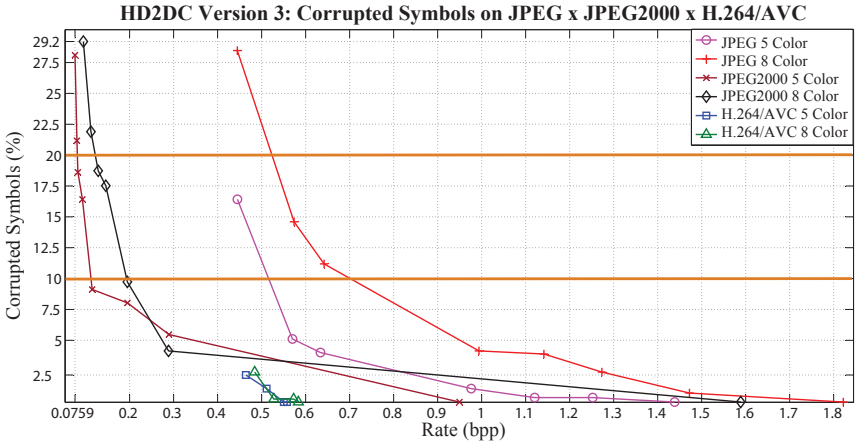


Fig. 16 Percentage of corrupted symbols of HD2DC Version 3 versus compression bitrate (0-2 bpp).

For the H.264/AVC compression algorithm, HD2DC-5 and HD2DC-8 can be decoded with compression bitrates higher than 0.3514 and 0.3627, respectively. These bitrates are the highest compression rates possible for this codec, which correspond to level 51 in H.264/AVC. The maximum percentages of corrupted symbols for these rates are 1.35% for the HD2DC-5 and 2.7% for HD2DC-8.

In summary, results show that, even when compressing HD2DC images, we obtained a good decoding performance. More specifically, a perfect HD2DC barcode can be decoded from a JPEG2k compressed image, for compression rates around 3.5 times lower than the compression ratios required for JPEG and H.264/AVC codecs.

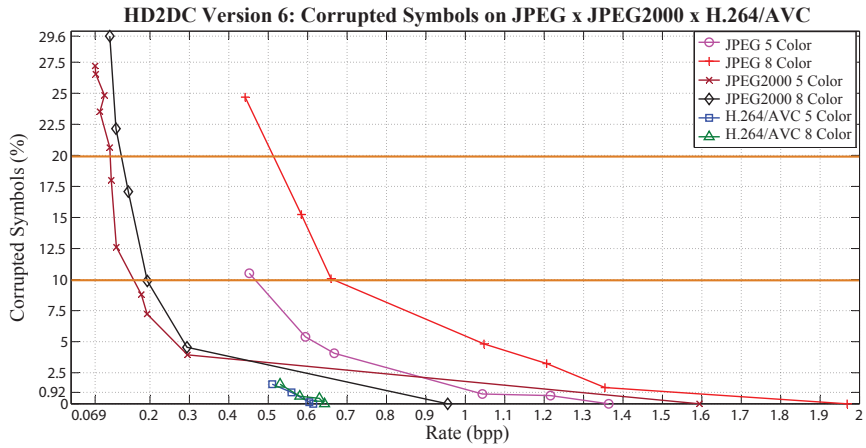


Fig. 17 Percentage of corrupted symbols of HD2DC Version 6 versus compression bitrate (0-2 bpp).

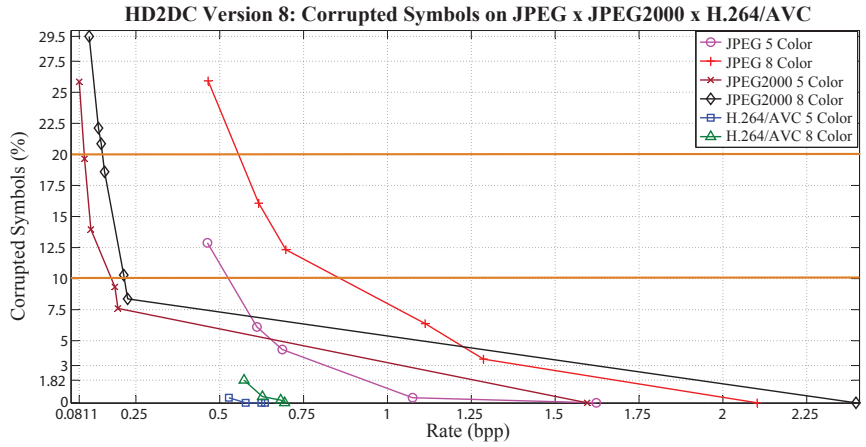


Fig. 18 Percentage of corrupted symbols of HD2DC Version 8 versus compression bitrate (0-2.5 bpp).

5 Conclusions

In this paper, we propose a new 2D colored barcode standard: the HD2DC. The proposed barcode can be generated with 5 or 8 colors, 8 different versions, and 3 error correction levels. HD2DC is capable of storing 2,906 bytes in an area of 2×2 cm, allowing an storage capacity of 4,687.09 bytes per in^2. Up to our knowledge, HD2DC is currently the highest data density 2D-barcode. This makes it adequate for cryptographic data validation, which requires a high amount of information to be coded and printed in very small areas.

Experimental tests were performed with all versions of HD2DC. In particular, we generated barcode images with a Low error correction level, printed them on regular paper, and captured them (using mobile phones) from distances varying from 7cm to 12cm. In all cases, the captured HD2DC barcode images were correctly decoded. We also tested the robustness of HD2DC to JPEG, JPEG2k and H.264/AVC compression algorithms. Results show that HD2DC can be correctly decoded even for very low bitrates. In summary, our experimental results indicate that the proposed HD2DC can be used for real multimedia applications, since it is robust to compression and print-scan distortions. We believe that even better results can be obtained by using more channel colors in the barcode structure.

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